

23m1049

I B.E. (EI & Mechanical)
 Class Test I October 2023
 Subject: 2EIRC4, Subject Name: Electrical Engineering

Duration: 70 min

Note: Attempt any four questions.

Max Marks: 20

Q1 A coil having a resistance of 10 ohms and inductance of 31.8 millihenry is connected to 230 V, 50 Hz supply calculates (a) the circuit current (b) power factor (c) power consumed. 5

Q2 Find the current flow through the battery in the circuit shown in figure (1). 5

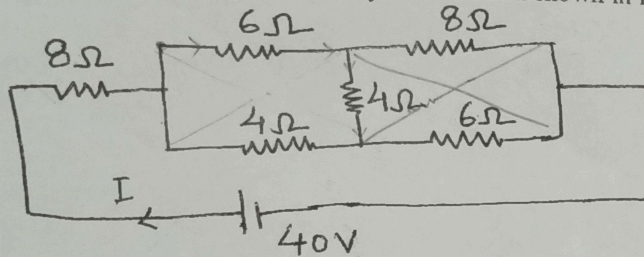


figure (1)

Q3 Use nodal analysis to determine the voltage V_1 and the current through 6 ohms resistance in the circuit shown in figure (2). 5

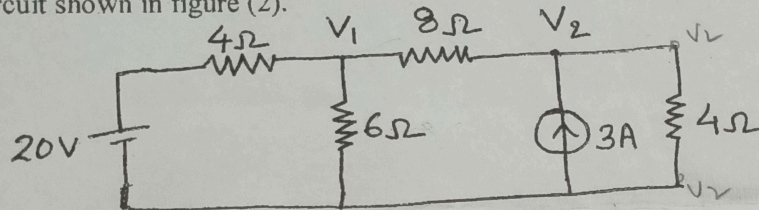


figure (2)

Q4 Find the current in branch AB of the given circuit figure (3) using thevenin's theorem. 5

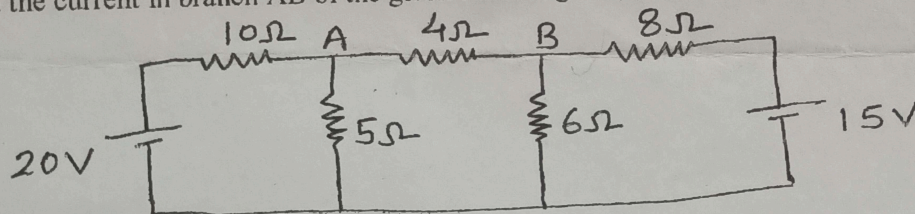


figure (3)

Q5 Calculate the current flowing I_1 , I_2 and I_3 in the circuit shown in figure (4). 5

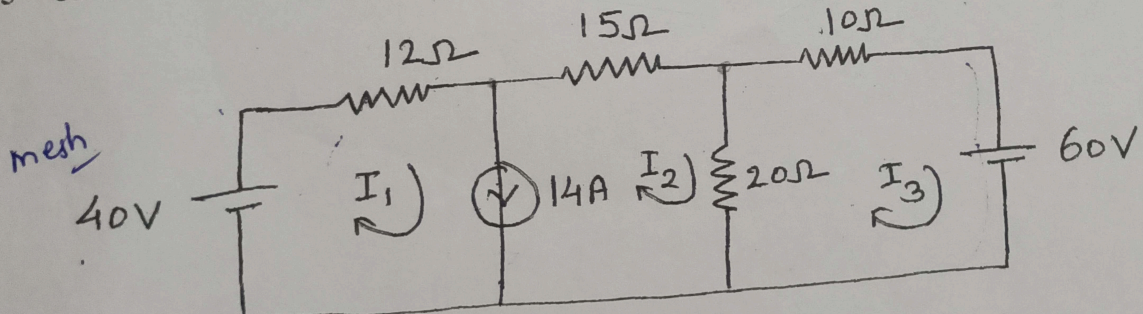


figure (4)